



NON-LINEAR DYNAMICS IN CHUA CIRCUIT

Agney K Rajeev

Integrated Physics Laboratory, School of Physical Sciences, NISER.

Introduction

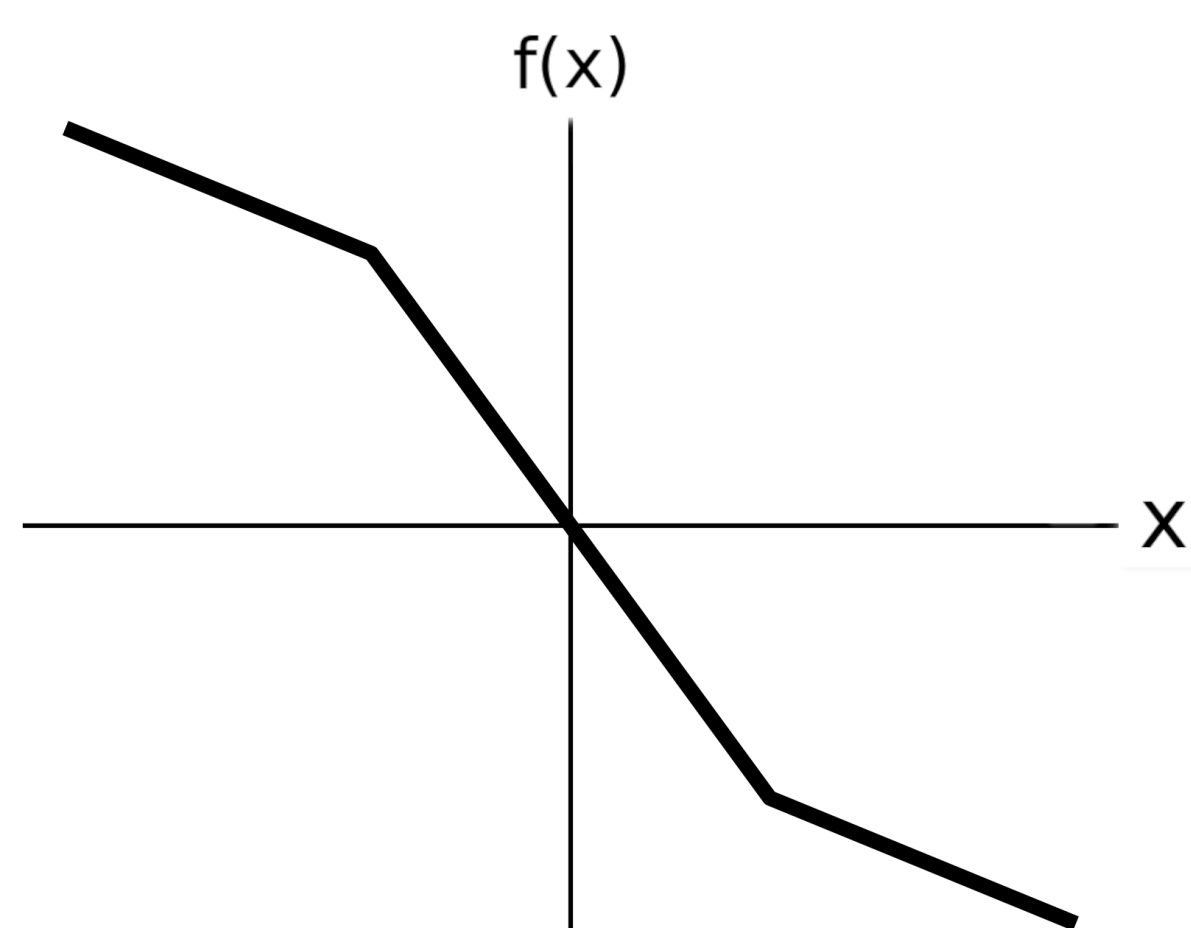
- **Nonlinear Dynamics** is the physics that studies systems governed by equations more complex than the linear $\mathbf{aX} + \mathbf{b}$ form
- Exhibit **deterministic chaos**, where their behaviour appears **unpredictable but are in fact not random** or unsystematic
- Electronic circuits provide a (1) **controlled & reproducible setting** and (2) **rapid response times** to model nonlinear phenomena
- **Chua Circuit** is the **simplest electronic circuit** which admits chaotic behaviour
- Invented in 1983 by **Leon O. Chua** and publicized in the paper “**A chaotic attractor from chua’s circuit**” by A. Matsumoto
- Displays numerous **Bifurcations** between **Fixed Point**, **Limit Cycles** and **Strange Attractors**

The Equations

$$\dot{x} = \alpha[y - x - f(x)]$$

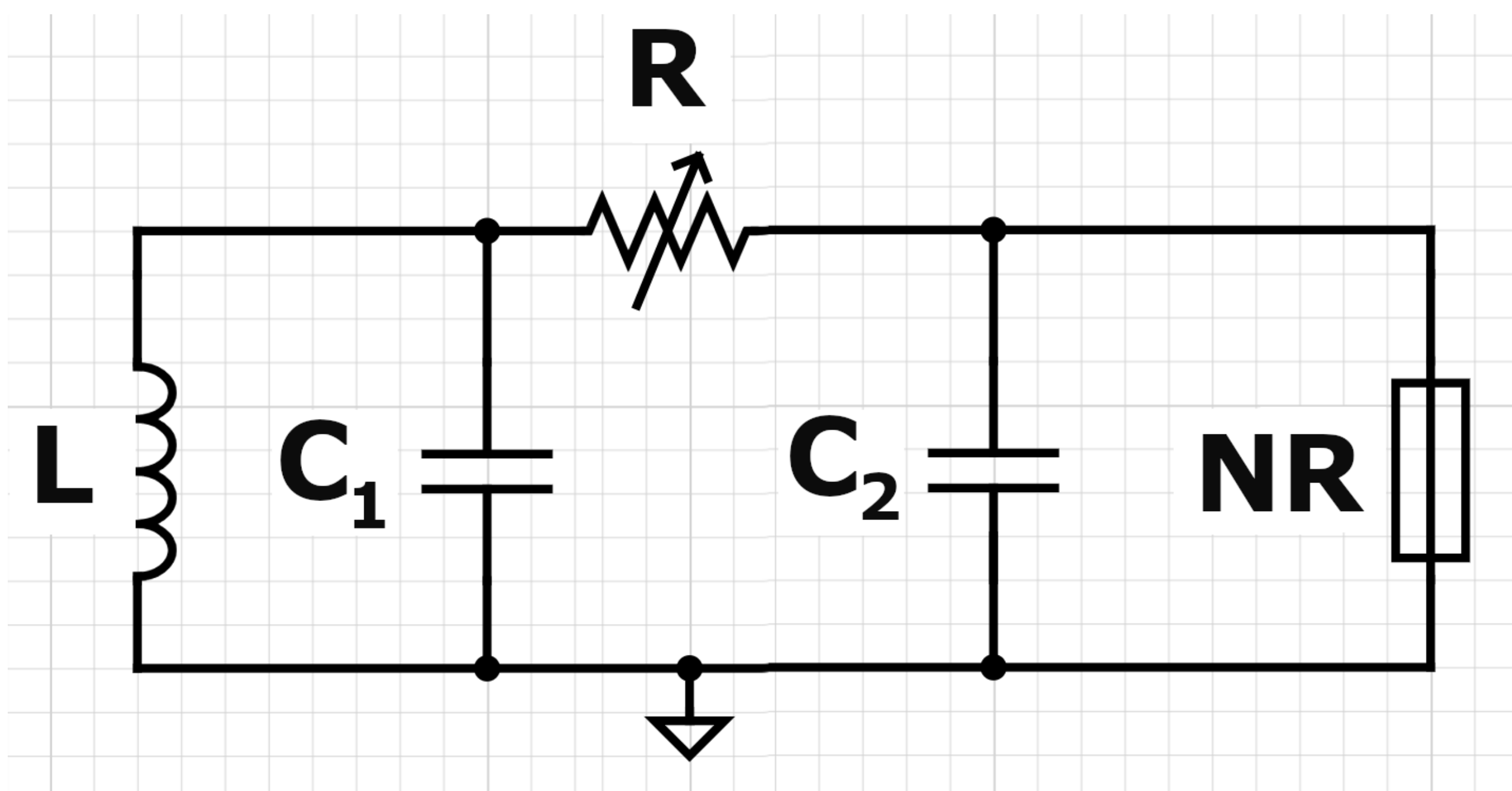
$$\dot{y} = x - y + z$$

$$\dot{z} = -\beta y$$

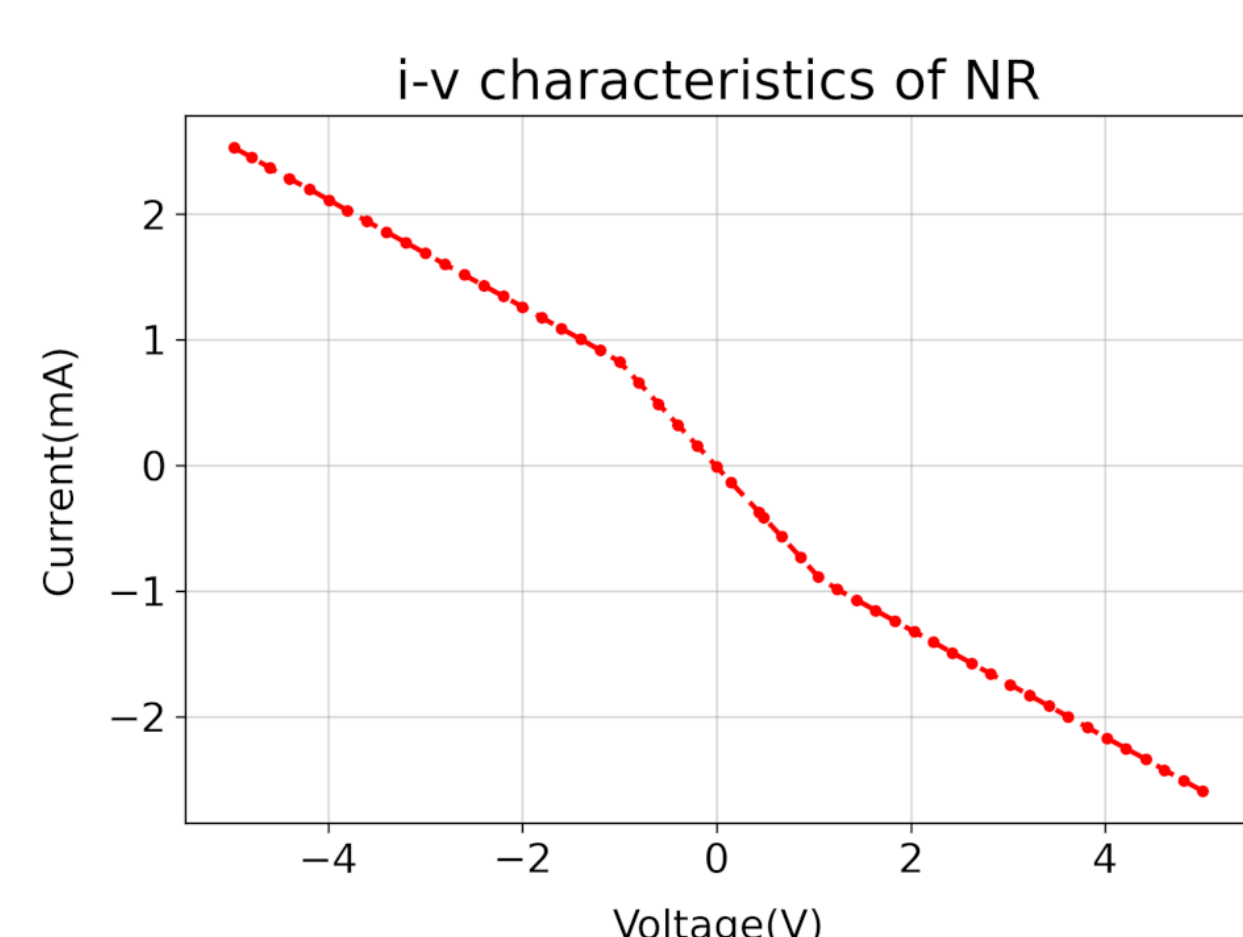
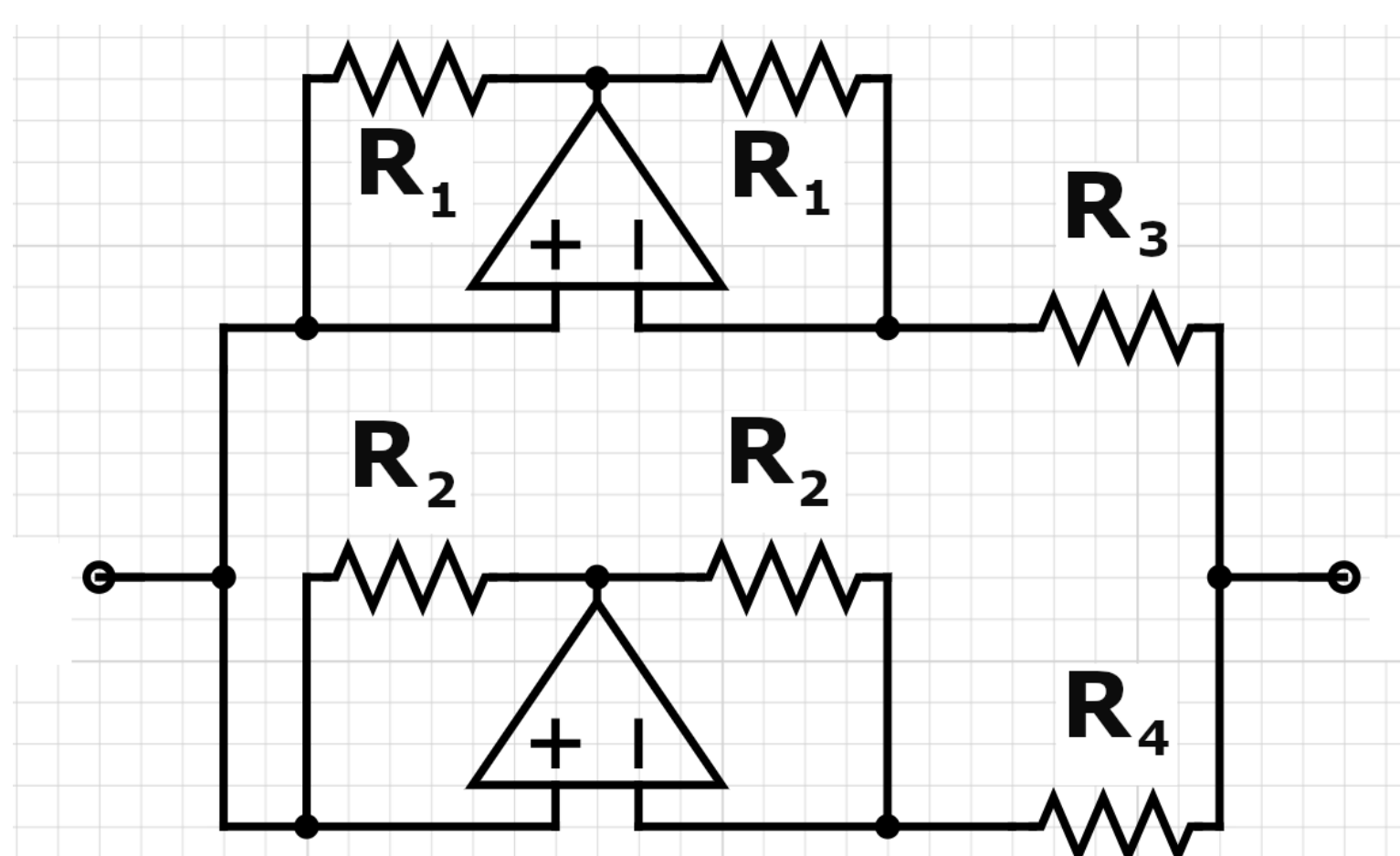


$f(x)$ is a **piecewise linear function** and the nonlinear component

Circuit Diagram



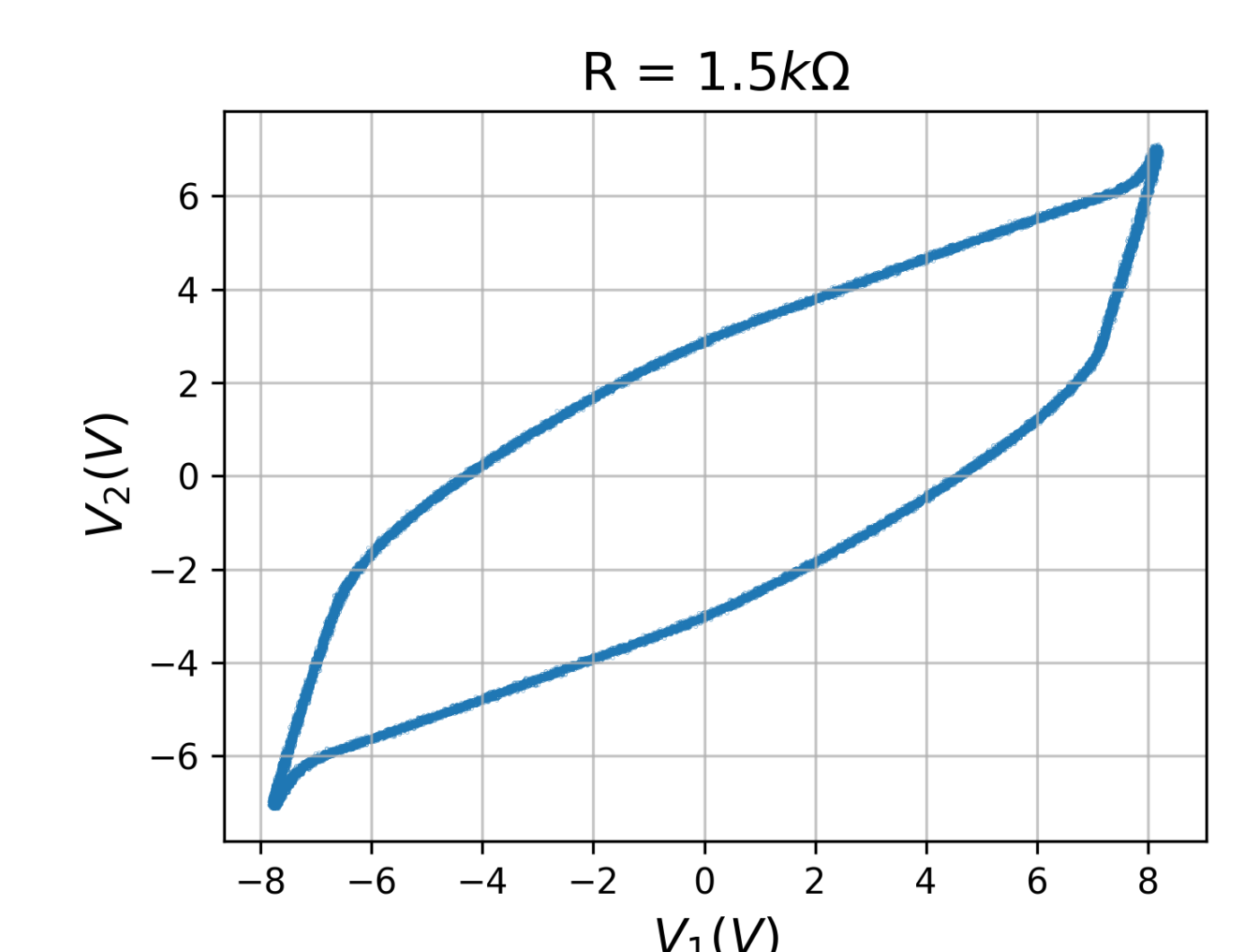
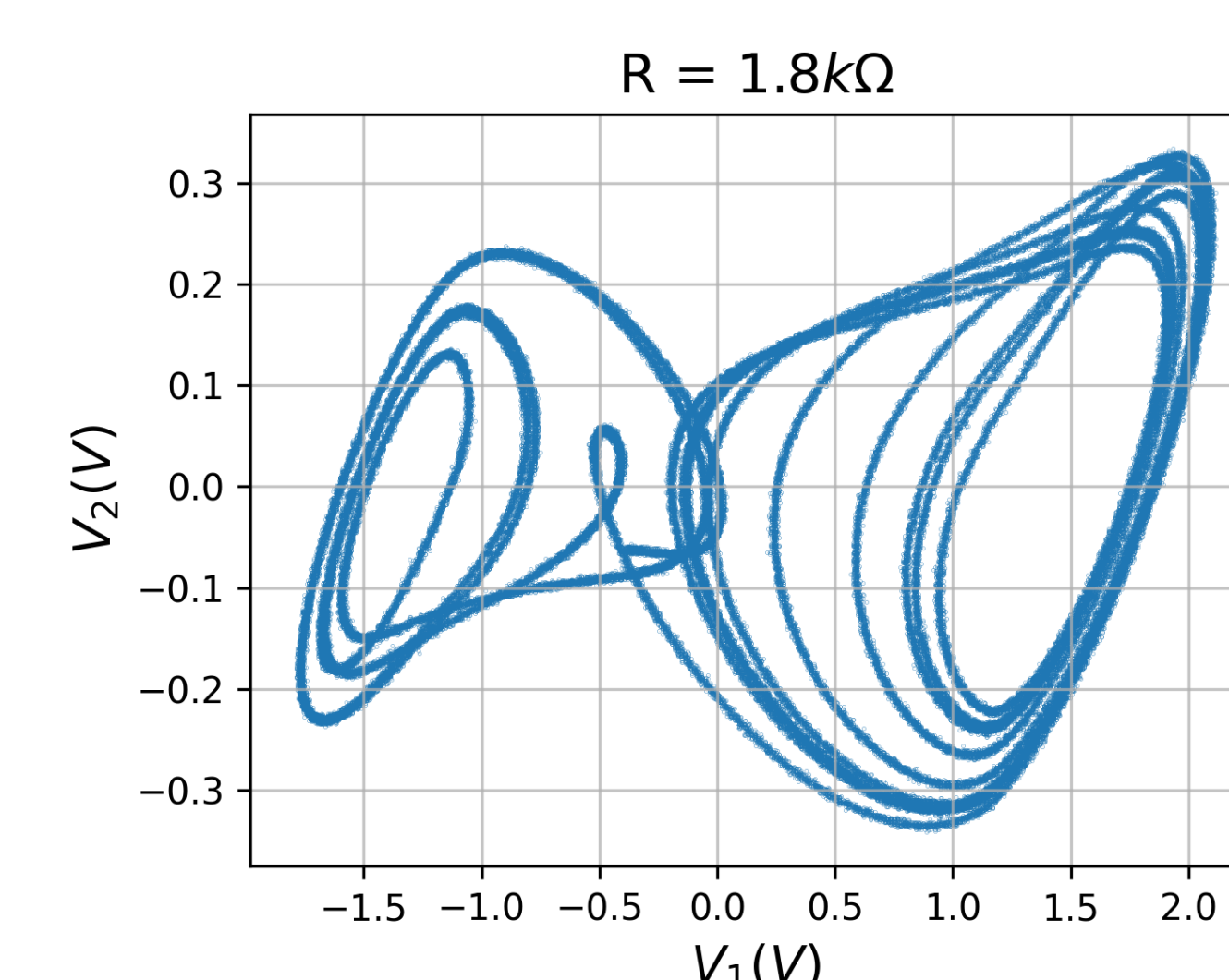
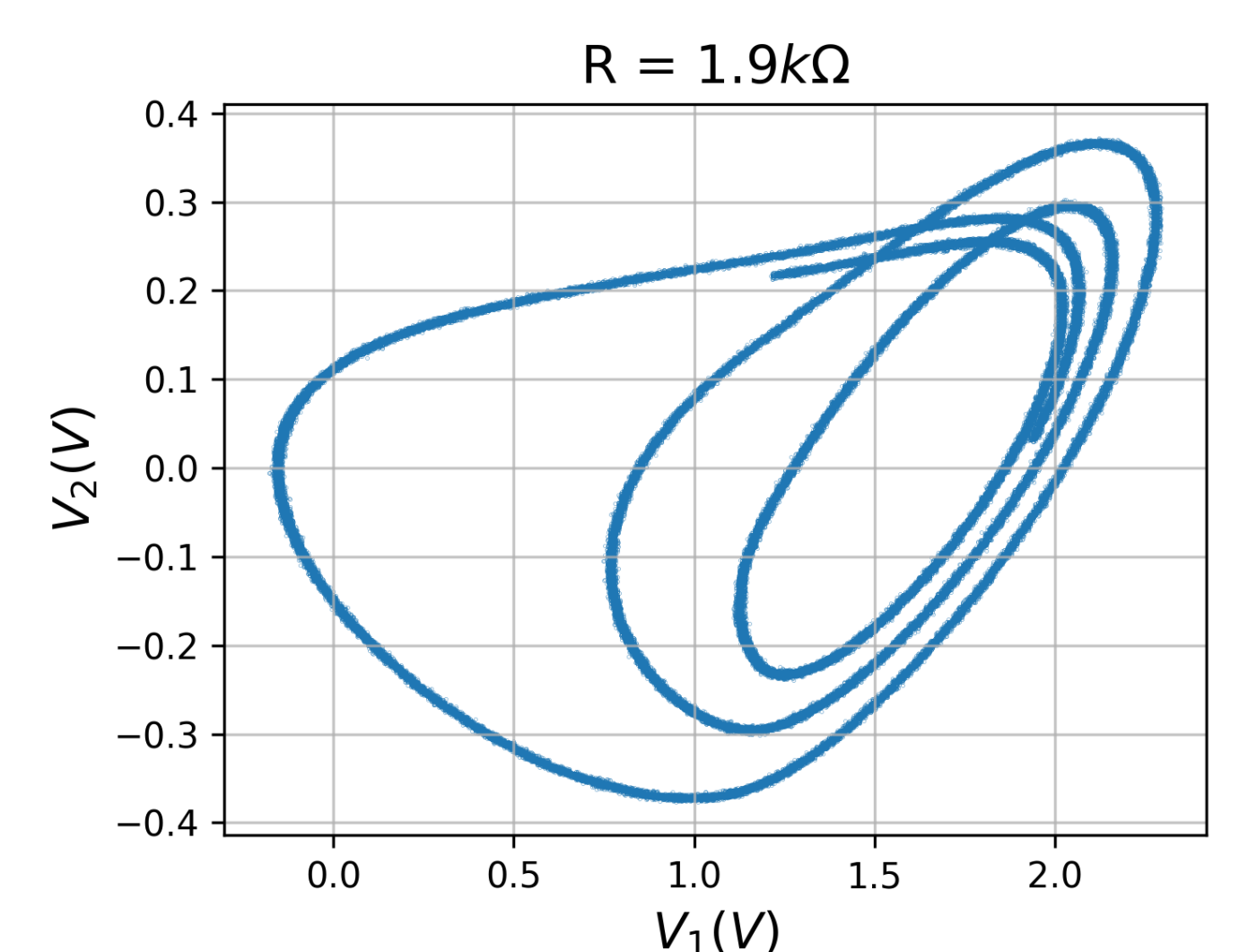
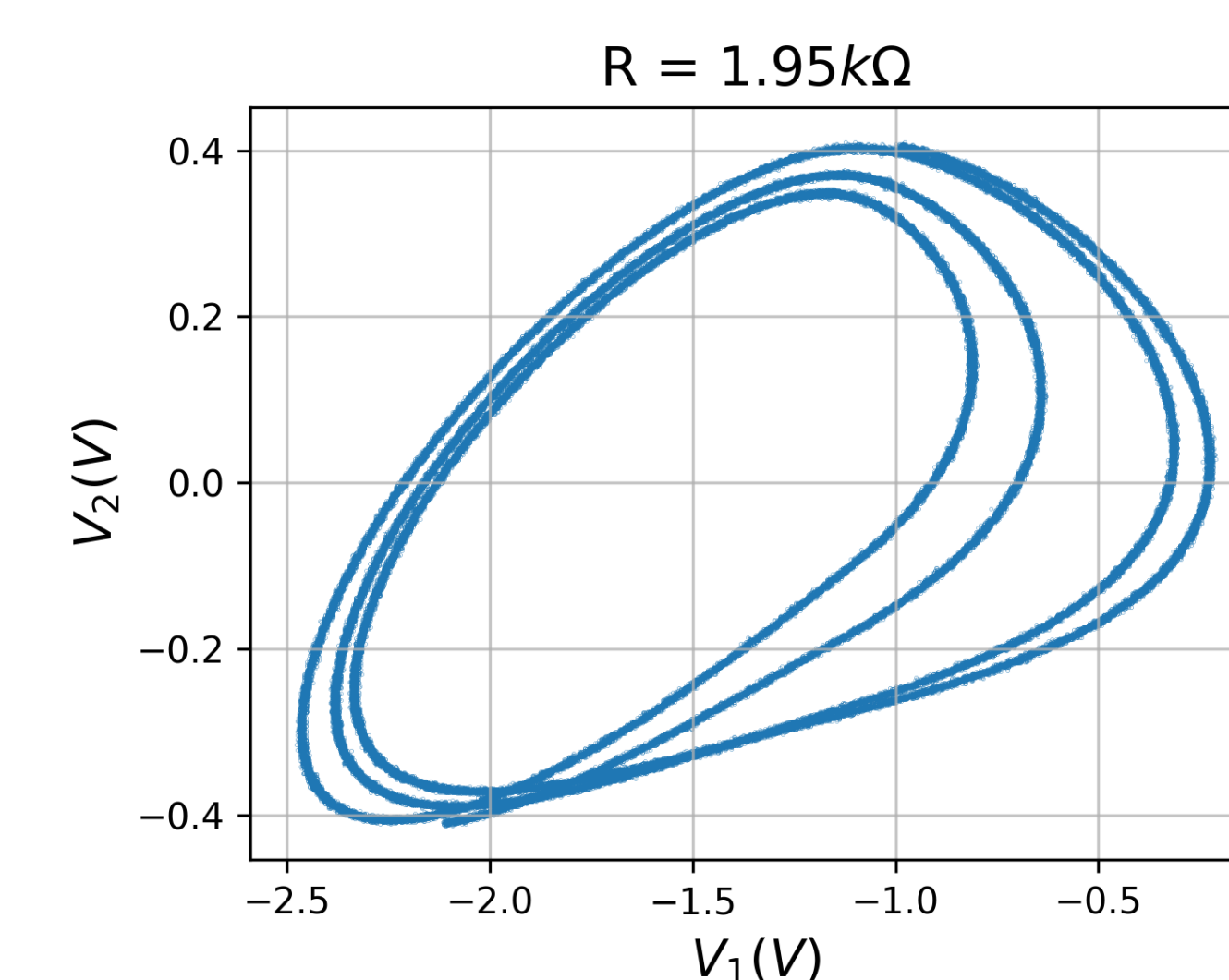
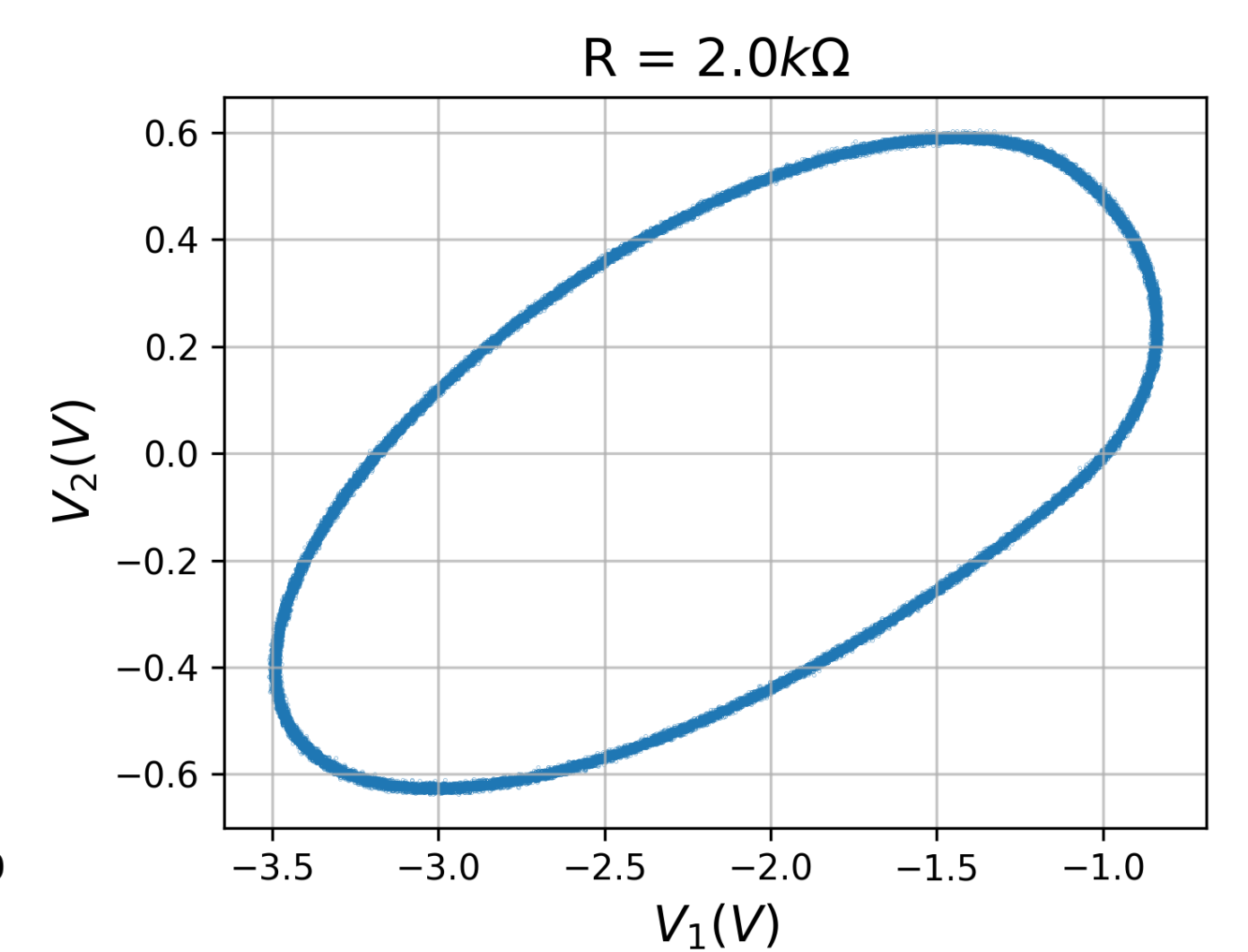
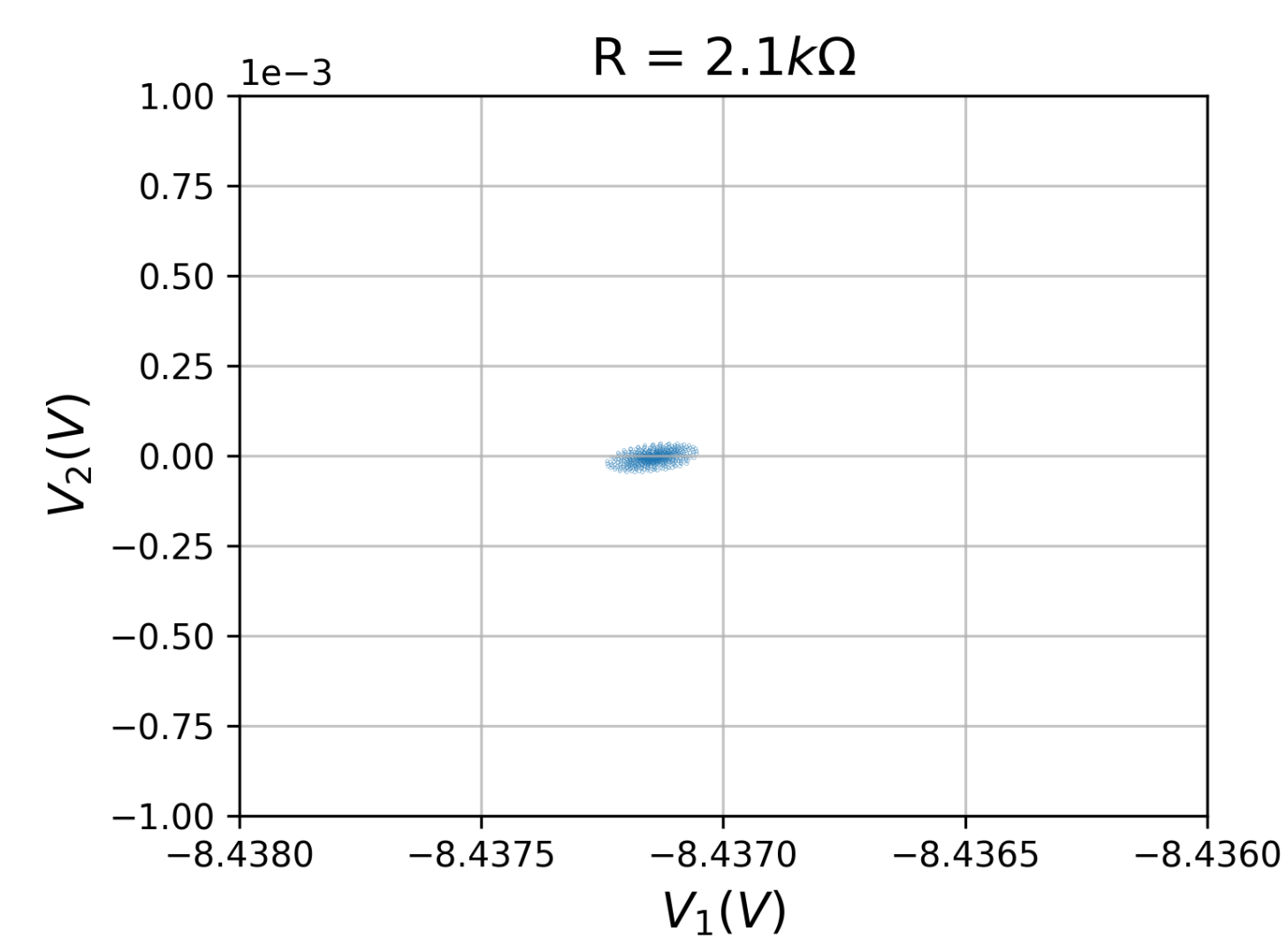
$$x = V_{C2} \quad y = V_{C1} \quad z = i_L R \quad \tau = t/RC_1 \quad f = i_{NR} R$$
$$\alpha = C_1/C_2 \quad \beta = R^2 C_1/L$$



Non-linear Resistor(NR) and its i-v characteristics

Results

$$L = 18mH \quad C_1 = 100nF \quad C_2 = 10nF \quad R_1 = 22k\Omega \quad R_2 = 220\Omega \quad R_3 = 3.3k\Omega \quad R_4 = 2.2k\Omega \quad OpAmp : TL082$$



From left to right then top to bottom, **Stable Fixed Point**, **1-Period Limit Cycle**, **Multi-Period Limit Cycle**, **Rosler Attractor**, **Double Scroll Attractor**, **Large Limit Cycle**

Applications

- Bifurcation Analysis
- Random Number Generation
- Secure Communications
- Neural Spiking Patterns

References

1. T. Matsumoto, “A chaotic attractor from chua’s circuit,” IEEE Transactions on Circuits and Systems, vol. 31, no. 12, pp. 1055–1058, 1984
2. M. P. Kennedy, “Robust OP amp realization of chua’s circuit,” Frequenz, vol. 46, Jan. 1992.